Fiscal Year:	FY 2008	**Task Last Updated:**	FY 05/21/2008
PI Name:	Bingham, Gail E Ph.D.		
Project Title:	Optimization of Root Zone Substrates for Reduced Gravity Experiments		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Advanced human support technologies		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	North Logan	**State:**	UT
Zip Code:	84341	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2000 Space Biology 00-HEDS-01
Start Date:	12/01/2001	**End Date:**	02/28/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 2/28/2009 per annual report info (5/2008) NOTE: End date corrected to 02/21/2008 per PI (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jones, Scott (Utah State University)		
Grant/Contract No.:	NAG 9-1284		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Gas diffusion in all root zones is limited by the air-filled pore space within the plant growth substrate. Since root mass is an actively respiring tissue, most plants require sufficient oxygen diffusion in their root environment to support their root respiration rate. When root zone oxygen becomes limiting, plant growth suffers. At 1g (on earth) gravity pulls the water through the root zone draining the macro pore spaces, allowing oxygen to diffuse to the roots. In mg, water is held in the macro pore spaces and gas diffusion into and out of the root zone can become a limiting factor for plant growth at lower volumetric water contents. The objective of this research is to determine the parameters required to optimize the plant growth root environment for reduced gravity environments so that future researchers can confidently tailor their root zone environment to satisfy their experimental needs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Removing gravitational effects from the measurements of oxygen and carbon dioxide diffusion in plant substrates has provided us with significant new insight to how these processes function on earth. We have been able to improve the modeling of water-soil interactions with the work to date. At the completion of this effort, we will be able to provide greenhouse and other earthbound high intensity plant growers with the tools to develop improved substrates and water management practices.
Task Progress:	We have completed the first ORZS (Optimization of Root-Zone Substrates) experiment, in which we concentrated on measuring the water retention characteristics of Profile, Turface, and Mix substrates. The hardware launched to the International Space Station (ISS) on May 12, 2007 and returned to Earth on February 20, 2008. When we compared oxygen diffusion measurements in microgravity with Earth-based data, preliminary results pointed to enhanced hysteresis in oxygen diffusion dependency on air-filled porosity in microgravity. These results indicate altered water distribution patterns relative to Earth-based measurements. We hypothesize that a critical air-filled pathway forms at a lower microgravity saturation level because of the absence of hydrostatic water distribution. We are now preparing a second ORZS flight experiment, from which, among other things, we will obtain data to verify this hypothesis. While attempting to model the diffusivity/air-filled porosity characteristic for Turface, Profile and Mix, we found that the previously advocated WLR model was inadequate to describe our data. We chose a new model that is flexible through introducing additional fitting parameters. We think this model will enable us to link physical changes in the microgravity environment to indicated changes in diffusivity/air-filled porosity behavior. The second ORZS unit was tested and certified with Profile and Turface at the same time as the first flight unit. The hardware has passed preship testing, and it is scheduled for delivery to Moscow on June 17. The hardware will then be tested, delivered, and ready to launch by the end of June on the August Progress.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Heinse R, Jones S, Bingham GE, Bugbee B. "Optimizing Straticulate Plant-Growth Media for Improved Root Zone Performance and Management." ASA-CSSA-SSSA 2007 International Annual Meetings, New Orleans, LA, November 4-8, 2007. Agronomy abstracts, 88. Agronomy abstracts, abstract #183-21, Am. Soc Agronomy, Madison, WI, 2007. p. 105. , Nov-2007
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Abstracts for Journals and Proceedings	Jones SB, Heinse R, Simunek J, Tuller M, Or D. "Numerical Modeling of Unsaturated Flows in Variable-gravity During Parabolic Flight" AGU Fall Meeting, San Francisco, CA, December 10-14, 2007. Eos Trans. AGU, 2007;88(52), Fall Meet. Suppl., Abstract H53E-1471. , Dec-2007
Abstracts for Journals and Proceedings	Jones SB, Heinse R, Or D, Poritz DH ,Topham ST, Podolsky IG, Bingham GE. "Oxygen Diffusion Measurements in Partially Saturated Porous Media Aboard the International Space Station." ASA-CSSA-SSSA 2007 International Annual Meetings, New Orleans, LA, November 4-8, 2007. Agronomy Abstracts. abstract 1004, session 183-11, 2007. , Nov-2007
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Articles in Peer-reviewed Journals	Or D, Tuller M, Jones SB. "Liquid Behavior in Partially-Saturated Porous Media under Variable Gravity." Soil Science Society of America Journal, under review, May 2008. , May-2008

Articles in Peer-reviewed Journals	Heinse R, Jones SB, Steinberg SL, Tuller M, Or D. "Measurements and modeling of variable gravity effects on water distribution and flow in unsaturated porous media." Vadose Zone Journal 2007 Nov;6(4):713-24. http://dx.doi.org/10.2136/vzj2006.0105 , Nov-2007
Significant Media Coverage	Jones SB, Topham S. "Root Zone Substrate Projects. " Utah Public Radio, August 2007. http://www.publicbroadcasting.net/upr/.jukebox?action=viewMedia&mediaId=621867 , Aug-2007